

ANJALIGUPTA RAGHAVENDRA

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SUMMARY

Data Engineer with 6+ years’ experience in analytics, ETL pipeline development, data validation, forecasting, and dashboard reporting using Python, SQL, and BI tools. Proven ability to transform large, multi-source datasets into analytics-ready assets to support decision-making. Complemented by hands-on experience applying machine learning, NLP, and Generative AI techniques including RAG-based systems using LangChain and FAISS to automate document analysis, scoring, and insight generation.

SKILLS

Languages	Python, SQL
Databases	SQL Server, PostgreSQL, BigQuery, Synapse
Data Engineering	ETL/ELT Pipeline Development, Data Warehousing, Star Schema Modeling, Fact & Dimension Tables, Batch Processing, Data Migration, CI/CD Implementation
Cloud Platforms	Microsoft Azure (ADF, ADLS Gen2, Synapse Analytics, Databricks, Azure DevOps, Azure Monitor, Key Vault), AWS (S3), Google Cloud Platform (BigQuery, Cloud Storage)
Data Validation & Governance	Data Reconciliation, Schema Validation, Business Rule Implementation, HIPAA Compliance, RBAC, Data Lineage Documentation
Python Libraries	pandas, NumPy, Matplotlib, ARIMA, Regression
Data Visualization & Reporting	Power BI, Tableau, Looker Studio
Tools & Development	Excel (Advanced Functions, Power Query, VBA), SSMS, Informatica, JIRA, Confluence
Retrieval-Augmented Generation (RAG)	LangChain, FAISS Vector Database, Document Chunking, Embeddings, LangGraph, Contextual Retrieval
Generative AI & LLMs	OpenAI GPT-5, Google Gemini 3, Prompt Engineering, LLM Evaluation, Automated Scoring Systems

EXPERIENCE

Optum Data Engineer	Jan 2024 – Present
- Designed and maintained scalable ETL pipelines using Azure Data Factory (ADF) to ingest Claims, Provider, Member, and Benefits data from SQL Server, SFTP, and REST APIs into Azure Data Lake Storage (ADLS Gen2) - Developed transformation workflows in Azure Databricks (PySpark) to cleanse, standardize, deduplicate, and enrich multi-million record healthcare datasets - Built dimensional data models (Fact & Dimension tables) in Azure Synapse Analytics (Dedicated SQL Pool) using star schema design to support finance, actuarial, and provider analytics reporting - Optimized Synapse performance using partitioning, clustered columnstore indexes, query tuning, and statistics management, improving query speed by 30%+ - Developed automated data validation and reconciliation checks using Python and SQL (row counts, null checks, schema validation, business rules) - Implemented CI/CD pipelines using Azure DevOps, ARM templates, and Git, enabling automated deployments across Dev, QA, and Prod environments - Built and deployed a multi-modal RAG medical equipment chatbot using Azure OpenAI, and FastAPI implementing prompt engineering experiments that improved technician equipment lookup time by 35% - Supported a Text-to-SQL Copilot using fine-tuned LLaMA2-7B (LoRA) to enable non-technical users to query enterprise healthcare data, reducing manual SQL development time by 65% - Configured monitoring, alerting, and secure data access using Azure Monitor, Log Analytics, RBAC, Azure AD, Managed Identities, Encryption, and Azure Key Vault, ensuring HIPAA compliance and improved SLA adherence - Collaborated with BI developers and analysts to deliver curated datasets for Power BI dashboards and enterprise reporting	
Environment: Azure Data Factory, Azure Databricks, PySpark, Azure Synapse Analytics, ADLS Gen2, Azure DevOps, Azure Monitor, Azure Key Vault, SQL Server, Power BI, Azure OpenAI, FastAPI, LLaMA2, ETL, Data Warehousing, CI/CD	

- Supported **regulatory and compliance data processing** for Truist Bank (escheatment) by building and maintaining **batch ETL pipelines** to transform high-volume **financial data into analytics-ready datasets**
- Developed complex SQL transformations (**CTE, Window Functions, Joins, Indexing**) and Python scripts to implement state-specific compliance rules, data validation, reconciliation, and regulatory mapping
- Developed **automated data validation and reconciliation frameworks** using to ensure accuracy of **balances, dormancy status, and reportable amounts** reducing manual validation effort by approx. 30%
- Built reusable **Python automation scripts** for **file ingestion, data cleansing, key-field standardization, and child-record categorization** prior to regulatory submission
- Utilized **AWS S3** for **secure file storage and cloud-based data movement** between on-prem systems and Enterprise Data Warehouse environments
- Created **Power BI dashboards and Excel (Power Query) reports** to monitor data quality metrics, ETL pipeline health, SLA adherence, and compliance reporting volumes
- Collaborated with **data engineers, and QA teams** to translate complex compliance requirements into ETL logic, **support UAT cycles**, perform **root cause analysis** on data discrepancies, and maintain clear documentation of **data lineage and transformation rules for audit and governance purposes**

Environment: Python (pandas, NumPy), SQL (SQL Server), AWS (S3), SSMS, Informatica, Excel (Power Query), Power BI, Git, Agile/Scrum, Enterprise Data Warehouses, Batch ETL Pipelines

GS INFOTECH

Data Analyst

Sep 2020 – Jun 2022

Junior Data Analyst

Jul 2018 – Aug 2020

- Analyzed 10M+ **POS and inventory transactions** across 100+ retail stores using SQL (**PostgreSQL, BigQuery**) to identify **revenue leakage, stock-out drivers, and category-level growth opportunities**
- Built **scalable analytical datasets** using **CTEs, window functions, joins, aggregations techniques** to calculate sell-through rate, inventory turnover, gross margin contribution, fill rate, and average selling price
- Led **demand forecasting** using Python (**pandas, NumPy, ARIMA, regression**), reducing forecast error (**MAPE**) by approx. 10% through seasonality and promotion impact modeling
- Performed **SKU-level aging** and **sales variability analysis** to optimize reorder points and safety stock thresholds, **reducing stock-out frequency** in fast-moving categories and **improving inventory turnover**
- Developed **Tableau dashboards** and automated **Excel reporting pipelines** using Python, reducing manual reporting effort and accelerating weekly business review readiness
- Leveraged **Google Cloud Platform (BigQuery, Cloud Storage)** to **migrate and scale retail analytics workloads**, improving query performance and **enabling near real-time KPI monitoring**

Environment: SQL (PostgreSQL, BigQuery), Python (pandas, NumPy, Matplotlib), Tableau, Excel, GCP (BigQuery, Cloud Storage), Time-Series Forecasting, Retail Analytics, KPI Modeling, Inventory Optimization, Demand Planning, Agile

PROJECTS

Research Grading Assistant Agent | *GenAI, RAG, LangChain, Gemini, FAISS* | [Link](#)

- Built an automated research paper grading assistant using Google Gemini 2.5 Pro, LangChain, and FAISS-based Retrieval-Augmented Generation (RAG), designed to evaluate academic papers in DOCX and PDF formats
- Engineered a dual-LLM pipeline (primary and fallback Gemini models) to ensure consistent grading accuracy across multiple components such as APA compliance, conclusion analysis, and thematic content evaluation
- Automated report generation with scores, section-wise breakdown, comment aggregation, and a 160-word summary using GenAI synthesis

Resume Analyzer Tool | *GenAI, LangChain, OpenAI, FAISS* | [Link](#)

- Developed an intelligent resume analysis application leveraging OpenAI GPT-4, and LangChain to match candidate resumes against job descriptions
- Designed an end-to-end pipeline for extracting and vectorizing resumes/JDs, enabling contextual retrieval and identification of key skills, strengths, gaps, and missing skills
- Enforced prompt engineering with a scoring system to evaluate technical proficiency, role fit, and domain relevance

EDUCATION

M.S in Information Science

May 2025

New Jersey Institute of Technology | GPA: 3.9/4.0

Newark, USA